

Python for data analysis

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3–9 October 2026 · ICMPA – UNESCO Chair, Université d'Abomey-Calavi, Cotonou, Benin

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Session's program

- 1) Essential tools to install
- 2) pandas library
- 3) xarray library
- 4) Model, in-situ and altimetry data

What's your OS?

Linux, Mac: all good, use terminal.

Windows: Let's install WSL2 to simulate Linux.

- Open a Power shell as administrators, and type:
 - `wsl --install -d Ubuntu`You can also install Ubuntu from the Microsoft store.
- Reboot your computer if asked so.
- Run Ubuntu from the start menu, and wait a moment.
- A **UNIX username** and **password** will be asked.
- From now on, enter all the commands in an Ubuntu terminal.
- To verify if you have ubuntu:
 - `uname -a` #Must mention 'Linux' and 'microsoft-standard-WSL2' on windows
 - `whoami` #You Linux username

Conda

Conda installs and isolates python libraries in a closed environment to avoid conflicts with the rest of your system.

- Download and install Miniconda:
 - `cd ~`
 - `wget https://repo.anaconda.com/archive/Anaconda3-2026.07-1-Linux-x86_64.sh`
 - `bash Miniconda3-latest-Linux-x86_64.sh`
- Accept the license, keep the default location (`~/miniconda`) and answer **yes** when you're asked to initialize.
- Close and reopen a terminal. Your command line should start by `(base)`, this is the name of your environment.
- Confirm by typing: `conda --version`

Create an environment

We want to create a dedicated environment for this school, where we'll install all the required python libraries. We just need to create the environment once. Then we can activate it every time we open a terminal.

- We'll install every package using the conda-forge channel. Taking Anaconda's `defaults` channel out of the configuration prevents mixed-channel conflicts.
 - `conda config --add channels conda-forge`
 - `conda config --remove channels defaults` #ignore "not found" if it isn't here
 - `conda config --set channel_priority strict`
- Clone the github repository and access it:
 - `git clone https://github.com/GGOSSS-Summer-School/GGOSSS-2026.git`
 - `cd GGOSSS-2026`
- Create the environment using the YAML file:
 - `conda env create -f ggoss2026_conda_env.yaml`
- To activate the environment:
 - `conda activate ggoss26`

You need to reactivate the environment each time you open a terminal!

Get the material of this session



Data:

- Download data from the drive:
https://drive.google.com/drive/folders/1a9_RhJpIXdrU36dJ1LmsJAXfxU0ij0F_?usp=sharing
- Or take it from a USB key circulated in the room.

Put them in the GGOSSS-2026/01-python-tutorial/data directory.

You can do it with the terminal:

```
cp -r PATH_TO_data_FOLDER ~/GGOSSS-2026/01-python-tutorial
```

Via drag and drop, you can copy the data folder, and paste it

To see the Linux files from Windows Explorer, go to: `\\ws1$\\Ubuntu\\home\\<linux-user>`

Notebooks:

- They are already in GGOSSS-2026/01-python-tutorial

Jupyter Notebook

Many of the sessions of the school are using Jupyter Notebooks. They combine code (Python) and text (markdown)

Jupyter is already installed in the environment we create.

Run jupyter lab in the working directory

- `cd GGOSSS-2026`
- `jupyter lab`

A page should open in your browser, you can see your directories on the left side.

If the page doesn't open, try opening of of the links written your terminal, which will look like the following text:

To access the server, open this file in a browser:

LINK 1 (starting with `file://`)

Or copy and paste one of these URLs:

LINK 2 (starting with `http://localhost`)

LINK 3 (starting with `http://`)

Look for `01-python-tutorial` in the list of directories on your jupyter lab page.

Questions?

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